Assignment 2

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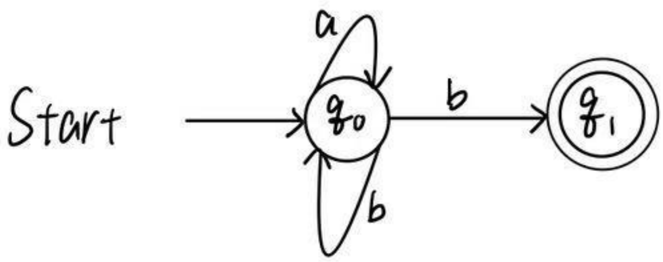
1 Required Exercises

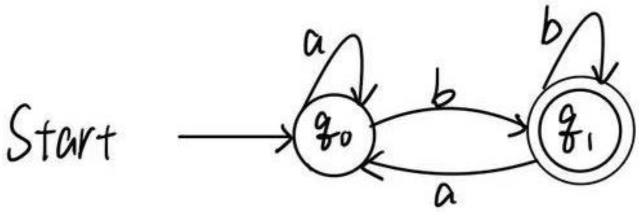
Exercise 1

1.

(1) NFA

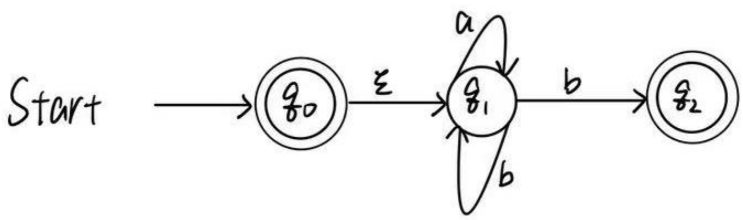


(2) DFA

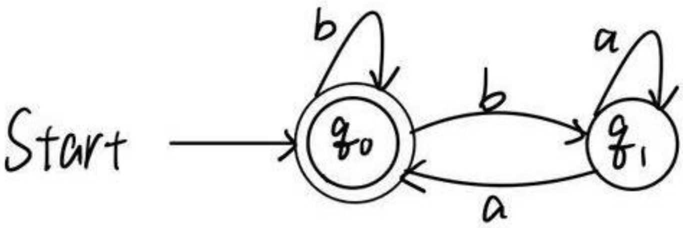


2.

(1) NFA

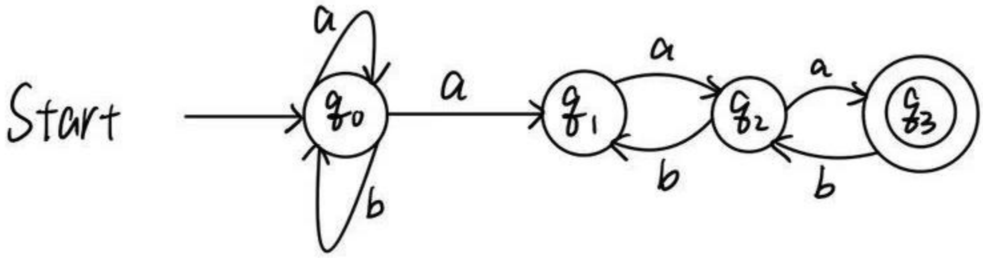


(2) DFA

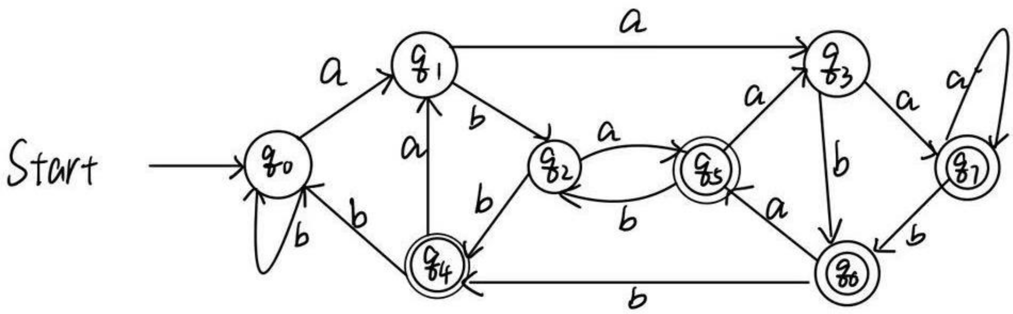


3.

(1) NFA

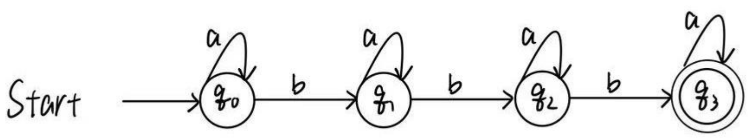


(2) DFA

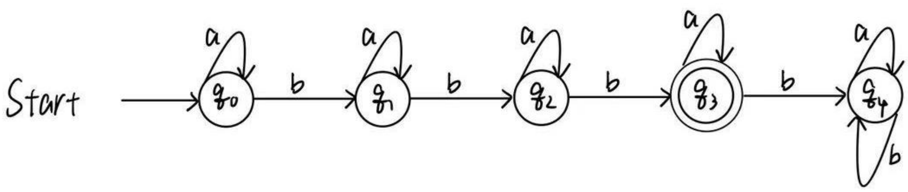


4.

(1) NFA



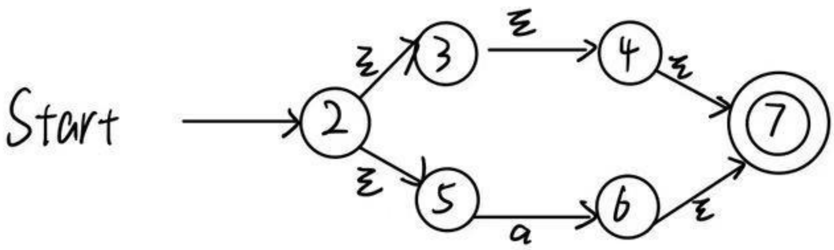
(2) DFA



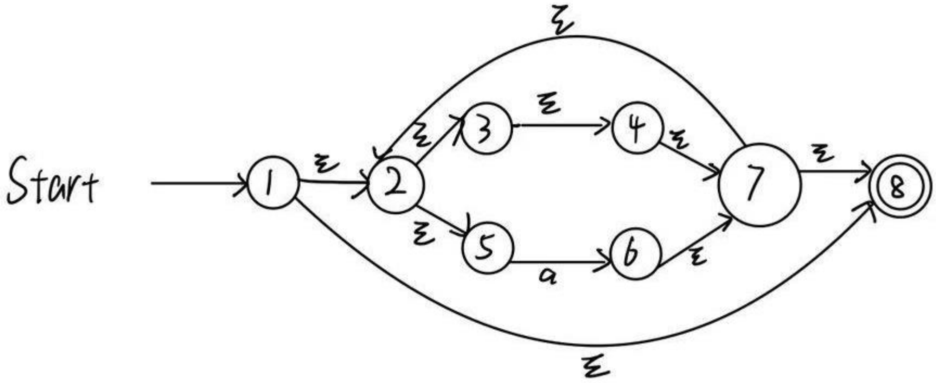
Exercise 2

1.

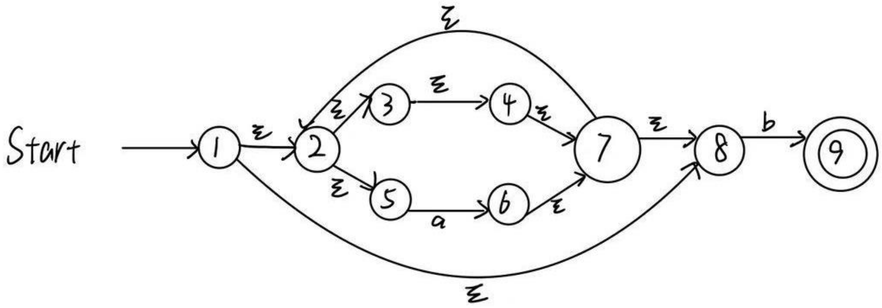
(1) Construct the NFA for $R_1 = \varepsilon | a$



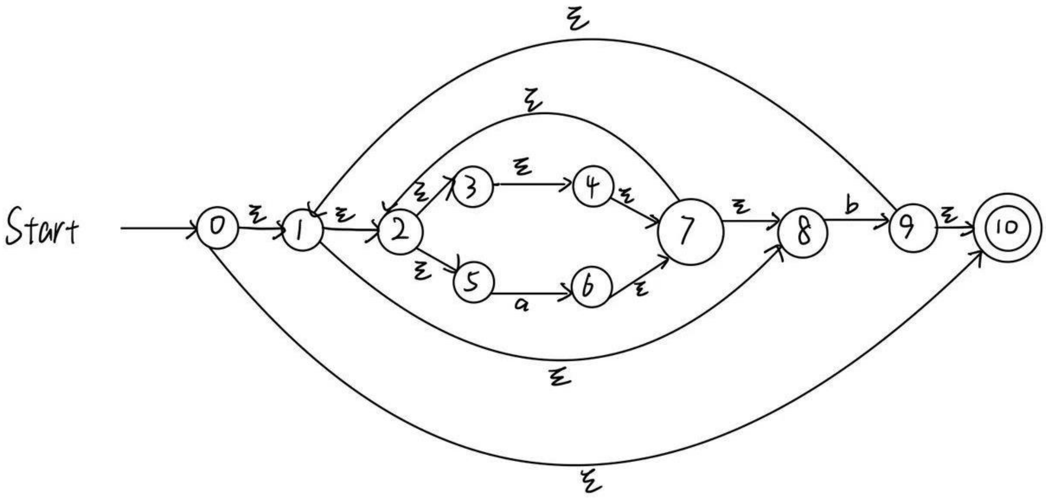
(2) Construct the NFA for $R_2 = (\varepsilon|a)^* = R_1^*$



(3) Construct the NFA for $R_3 = (\varepsilon|a)^* b = R_2 b$

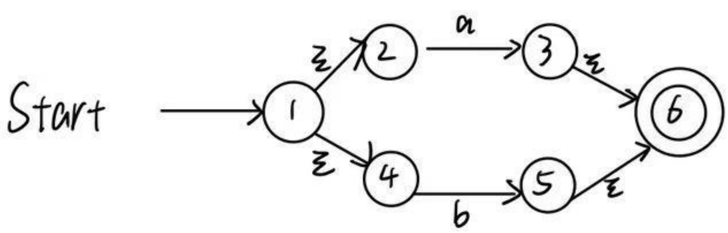


(4) Construct the NFA for $((\varepsilon|a)^* b)^* = R_3^*$

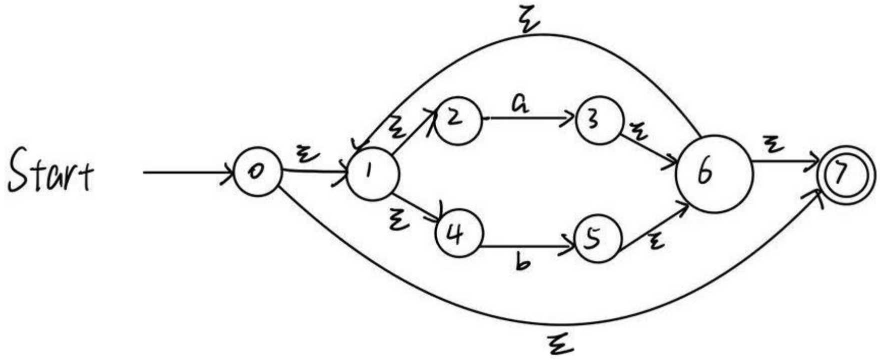


2.

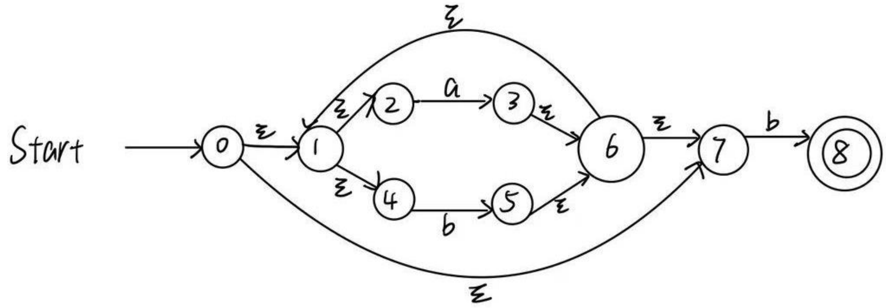
(1) Construct the NFA for $R_1 = a|b$.



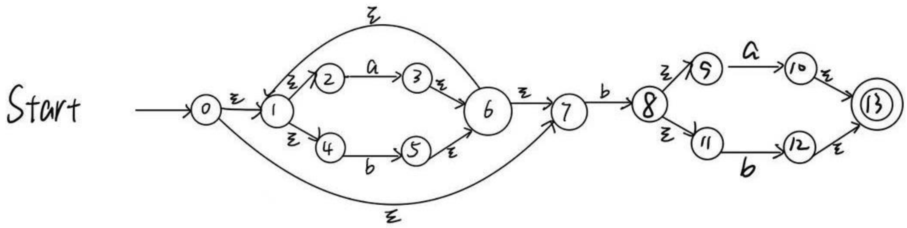
(2) Construct the NFA for $R_2 = (a|b)^* = R_1^*$.



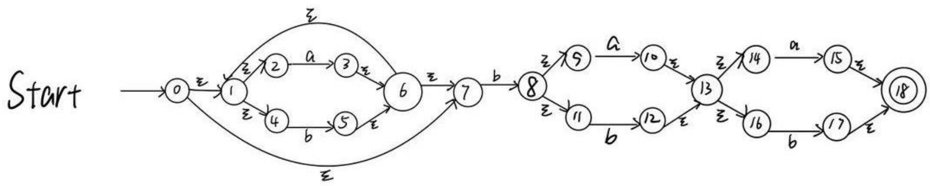
(3) Construct the NFA for $R_3 = (a|b)^*a = R_2a$.



(4) Construct the NFA for $R_4 = (a|b)^*a(a|b) = R_3R_1$.

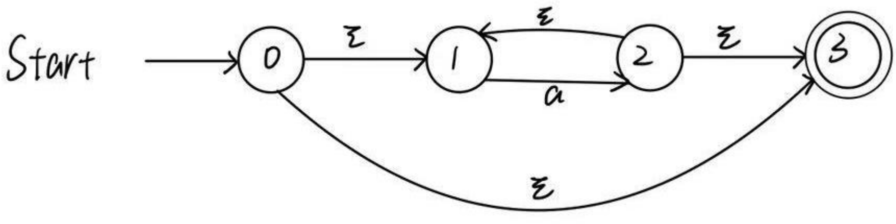


(5) Construct the NFA for $(a|b)^*a(a|b)(a|b) = R_4R_1$.

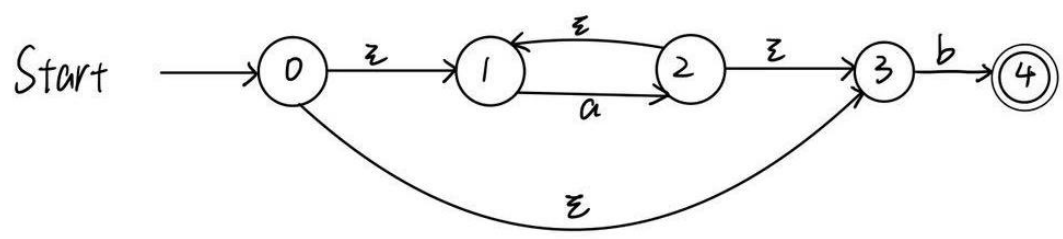


3.

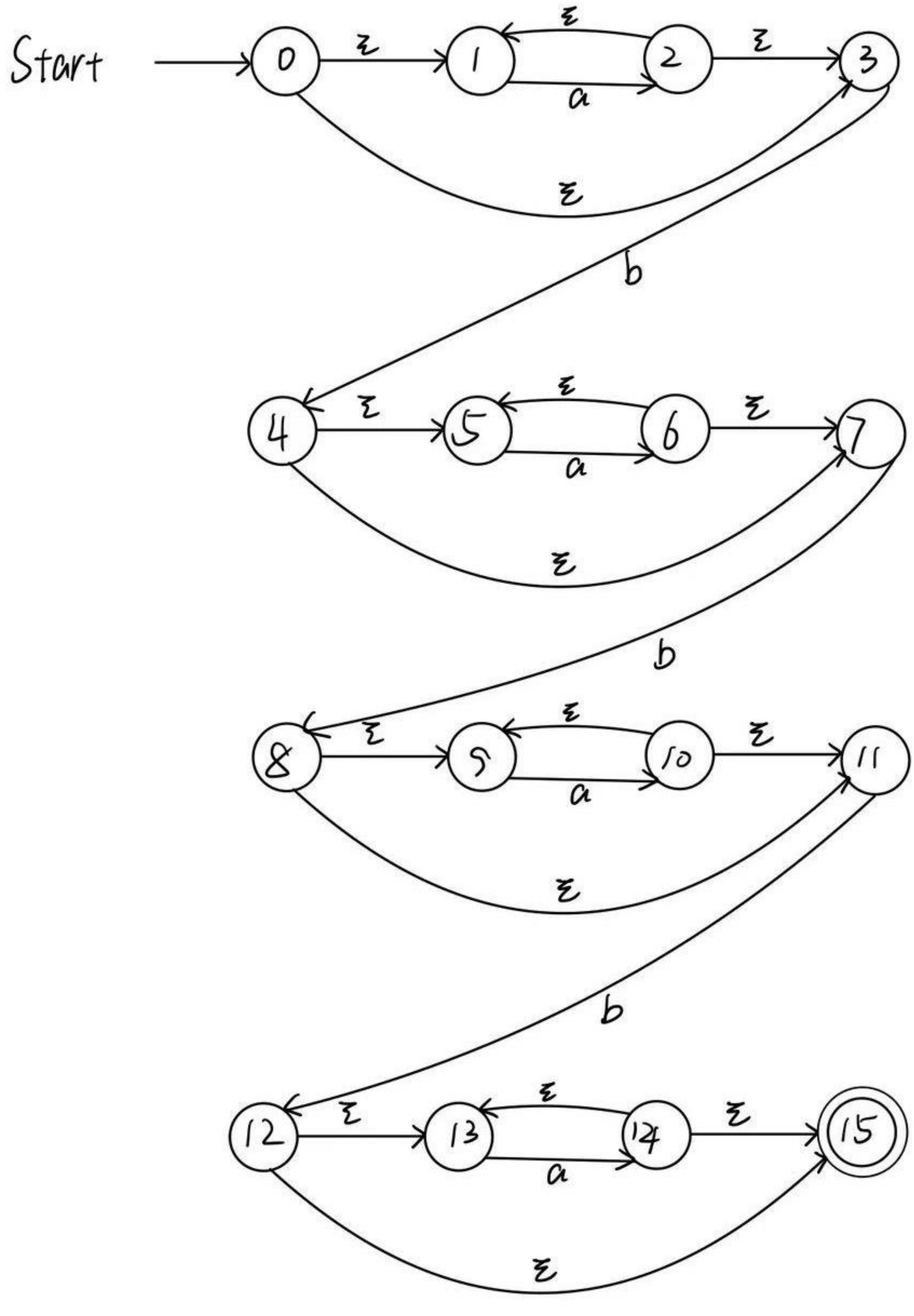
(1) Construct the NFA for $R_1 = a^*$.



(2) Construct the NFA for $R_2 = a^*b = R_1b$.



(3) Construct the NFA for $a^*ba^*ba^*ba^* = R_2R_2R_2R_1$.



Exercise 3

1.

$A = \varepsilon - closure(\{0\}) = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 10\}$.

$B = \delta_D(A, a) = \varepsilon - closure(\{6\}) = \{2, 3, 4, 5, 6, 7, 8\}$.

$C = \delta_D(A, b) = \varepsilon - closure(\{9\}) = \{1, 2, 3, 4, 5, 7, 8, 9, 10\}$.

$D = \delta_D(B, a) = \varepsilon - closure(\{6\}) = B$.

$E = \delta_D(B, b) = \varepsilon - closure(\{9\}) = C$.

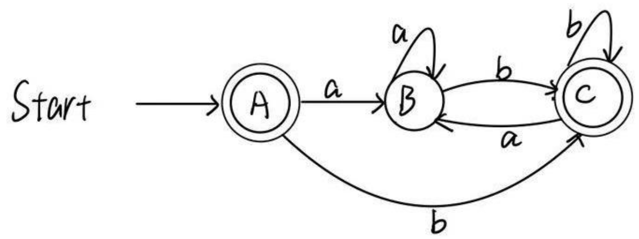
$F = \delta_D(C, a) = \varepsilon - closure(\{6\}) = B$.

$G = \delta_D(C, b) = \varepsilon - closure(\{9\}) = C$.

The transition table of the DFA (Starting state is A and finite states are {A, C}):

NFA STATE	DFA STATE	a	b
$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 10\}$	A	B	C
$\{2, 3, 4, 5, 6, 7, 8\}$	B	B	C
$\{1, 2, 3, 4, 5, 7, 8, 9, 10\}$	C	B	C

The transition diagram:



2.

$A = \varepsilon - closure(\{0\}) = \{0, 1, 2, 4, 7\}$.

$B = \delta_D(A, a) = \varepsilon - closure(\{3, 8\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 11\}$.

$C = \delta_D(A, b) = \varepsilon - closure(\{5\}) = \{1, 2, 4, 5, 6, 7\}$.

$D = \delta_D(B, a) = \varepsilon - closure(\{3, 8, 10\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 11, 13, 14, 16\}$.

$E = \delta_D(B, b) = \varepsilon - closure(\{5, 12\}) = \{1, 2, 4, 5, 6, 7, 12, 13, 14, 16\}$.

$F = \delta_D(C, a) = \varepsilon - closure(\{3, 8\}) = B$.

$G = \delta_D(C, b) = \varepsilon - closure(\{5\}) = C$.

$H = \delta_D(D, a) = \varepsilon - closure(\{3, 8, 10, 15\})$

$= \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 18\}$.

$I = \delta_D(D, b) = \varepsilon - closure(\{5, 12, 17\}) = \{1, 2, 4, 5, 6, 7, 12, 13, 14, 16, 17, 18\}$.

$J = \delta_D(E, a) = \varepsilon - closure(\{3, 8, 15\}) = \{1, 2, 3, 4, 6, 7, 8, 9, 11, 13, 14, 15, 18\}$.

$K = \delta_D(E, b) = \varepsilon - closure(\{5, 17\}) = \{1, 2, 4, 5, 6, 7, 17, 18\}$.

$L = \delta_D(H, a) = \varepsilon - closure(\{3, 8, 10, 15\}) = H$.

$M = \delta_D(H, b) = \varepsilon - closure(\{5, 12, 17\}) = I$.

$N = \delta_D(I, a) = \varepsilon - closure(\{3, 8, 15\}) = J$.

$O = \delta_D(I, b) = \varepsilon - closure(\{5, 17\}) = K$.

$P = \delta_D(J, a) = \varepsilon - closure(\{3, 8, 10\}) = D$.

$Q = \delta_D(J, b) = \varepsilon - closure(\{5, 12\}) = E$.

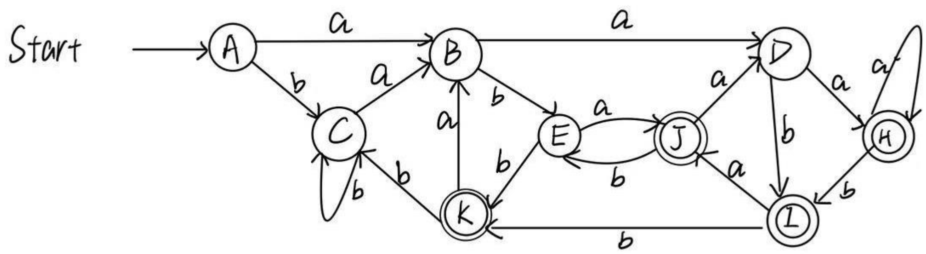
$R = \delta_D(K, a) = \varepsilon - closure(\{3, 8\}) = B$.

$S = \delta_D(K, b) = \varepsilon - closure(\{5\}) = C$.

The transition table of the DFA (Starting state is A and finite states are {H, I, J, K}):

NFA STATE	DFA STATE	a	b
$\{0, 1, 2, 4, 7\}$	A	B	C
$\{1, 2, 3, 4, 6, 7, 8, 9, 11\}$	B	D	E
$\{1, 2, 4, 5, 6, 7\}$	C	B	C
$\{1, 2, 3, 4, 6, 7, 8, 9, 11, 13, 14, 16\}$	D	H	I
$\{1, 2, 4, 5, 6, 7, 12, 13, 14, 16\}$	E	J	K
$\{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 18\}$	H	H	I
$\{1, 2, 4, 5, 6, 7, 12, 13, 14, 16, 17, 18\}$	I	J	K
$\{1, 2, 3, 4, 6, 7, 8, 9, 11, 13, 14, 15, 18\}$	J	D	E
$\{1, 2, 4, 5, 6, 7, 17, 18\}$	K	B	C

The transition diagram:



3.

$A = \varepsilon - closure(\{0\}) = \{0, 1, 3\}$.

$B = \delta_D(A, a) = \varepsilon - closure(\{2\}) = \{1, 2, 3\}$.

$C = \delta_D(A, b) = \varepsilon - closure(\{4\}) = \{4, 5, 7\}$.

$D = \delta_D(B, a) = \varepsilon - closure(\{2\}) = B$.

$E = \delta_D(B, b) = \varepsilon - closure(\{4\}) = C$.

$F = \delta_D(C, a) = \varepsilon - closure(\{6\}) = \{5, 6, 7\}$.

$G = \delta_D(C, b) = \varepsilon - closure(\{8\}) = \{8, 9, 11\}.$

$H = \delta_D(F, a) = \varepsilon - closure(\{6\}) = F.$

$I = \delta_D(F, b) = \varepsilon - closure(\{8\}) = G.$

$J = \delta_D(G, a) = \varepsilon - closure(\{10\}) = \{9, 10, 11\}.$

$K = \delta_D(G, b) = \varepsilon - closure(\{12\}) = \{12, 13, 15\}.$

$L = \delta_D(J, a) = \varepsilon - closure(\{10\}) = J.$

$M = \delta_D(J, b) = \varepsilon - closure(\{12\}) = K.$

$N = \delta_D(K, a) = \varepsilon - closure(\{14\}) = \{13, 14, 15\}.$

$O = \delta_D(K, b) = \emptyset.$

$P = \delta_D(N, a) = \varepsilon - closure(\{14\}) = N.$

$Q = \delta_D(N, b) = \emptyset = O.$

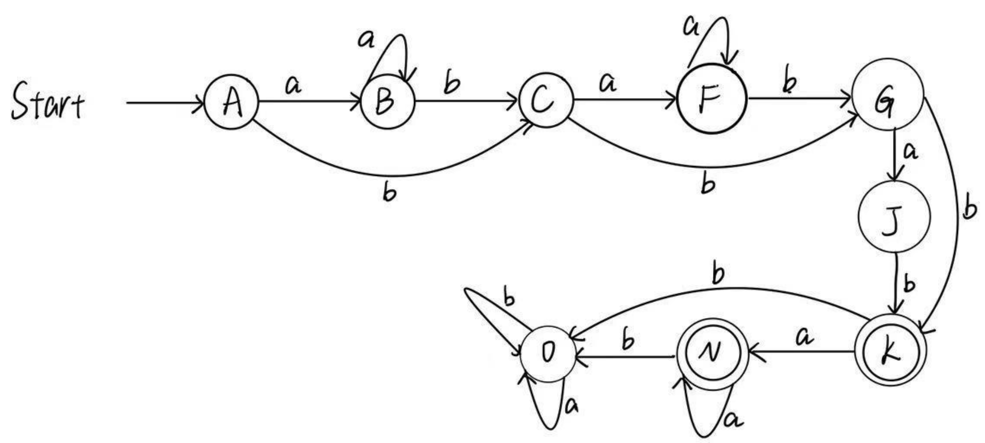
$R = \delta_D(N, b) = \emptyset = O.$

$S = \delta_D(N, b) = \emptyset = O.$

The transition table of the DFA (Strating state is A and finite states are {K, N}):

NFA STATE	DFA STATE	a	b
$\{0, 1, 3\}$	A	B	C
$\{1, 2, 3\}$	B	B	C
$\{4, 5, 7\}$	C	F	G
$\{5, 6, 7\}$	F	F	G
$\{8, 9, 11\}$	G	J	K
$\{9, 10, 11\}$	J	J	K
$\{12, 13, 15\}$	K	N	O
$\{13, 14, 15\}$	N	N	O
$\emptyset$	O	O	O

The transition diagram:



2 Optional Exercises

Exercise 1

Regular expression 2:

(1) Initial partition: $G_1 = \{A, B, C, D, E\}, G_2 = \{H, I, J, K\}$.

(2) G_1: b splits it into two groups $G_3 = \{A, B, C, D\}, G_4 = \{E\}$.

(3) G_3: a splits it into two groups $G_5 = \{A, B, C\}, G_6 = \{D\}$.

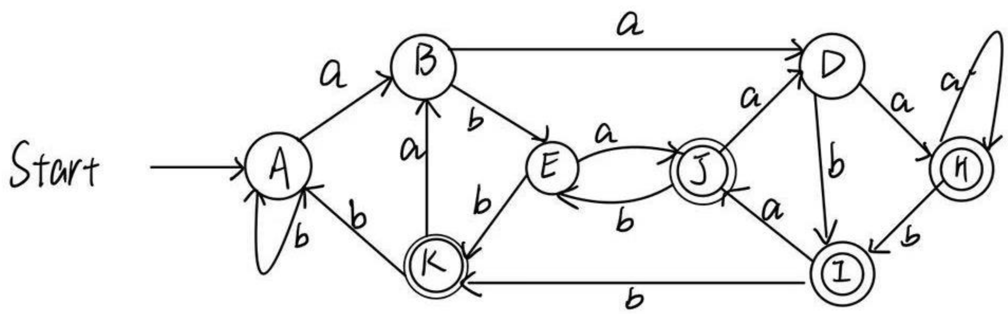
(4) G_5: b splits it into two groups $G_7 = \{A, C\}, G_8 = \{B\}$.

(5) G_2: a splits it into two groups $G_9 = \{H, I, J\}$ $G_{10} = \{K\}$.

(6) G_9: b splits it into two groups $G_{11} = \{H, J\}, G_{12} = \{I\}$.

(7) G_{11}: a splits it into two groups $G_{13} = \{H\}, G_{14} = \{J\}$.

(8) Picking A, B, D, E, H, I, J, K as the representatives to construct the minimum-state DFA.



Regular expression 3:

(1) Initial partition: $G_1 = \{A, B, C, F, G, J, O\}, G_2 = \{K, N\}$.

(2) G_1: b splits it into two groups $G_3 = \{A, B, C, F, O\}, G_4 = \{G, J\}$.

(3) G_3: b splits it into two groups $G_5 = \{A, B, O\}, G_6 = \{C, F\}$.

(4) G_5: b splits it into two groups $G_7 = \{A, B\}, G_8 = \{O\}$.

(5) Picking A, C, G, K, O as the representatives to construct the minimum-state DFA.